

Maschinelles Lernen für die Chemie - Exercise 2

Bayes' Theorem, Data Types and Regression
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Pandas Methods

`pandas.DataFrame.value_counts()`

- Count the number of occurrences of each unique value in a DataFrame

`pandas.DataFrame.info()`

- Print summary of the DataFrame
- Similar to `pandas.DataFrame.describe()`

`pandas.DataFrame.copy()`

- Copy selected sub-DataFrames (e. g. columns) to a new DataFrame variable
- Changes in the copy will not influence the copy (default behavior)

Exercise

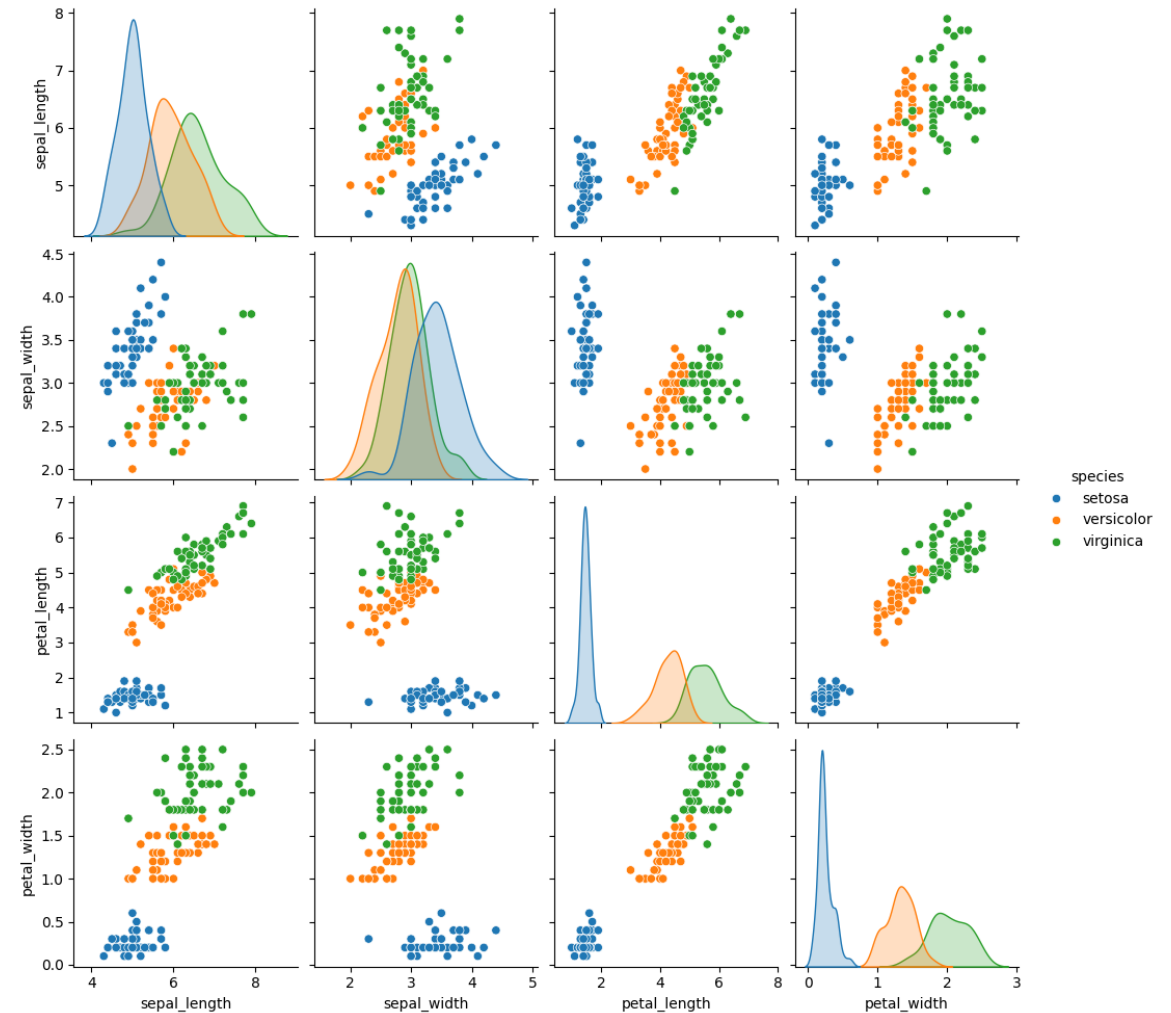
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Pair Plot

`sns.pairplot(DataFrame, hue="labels")`

- Easy way to look at obvious correlations between different features

Exercise
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Scaling Input Data

Scikit-learn StandardScaler

- Scale data to a mean value of $\mu = 0$ and standard deviation of $\sigma = 1$

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma}$$

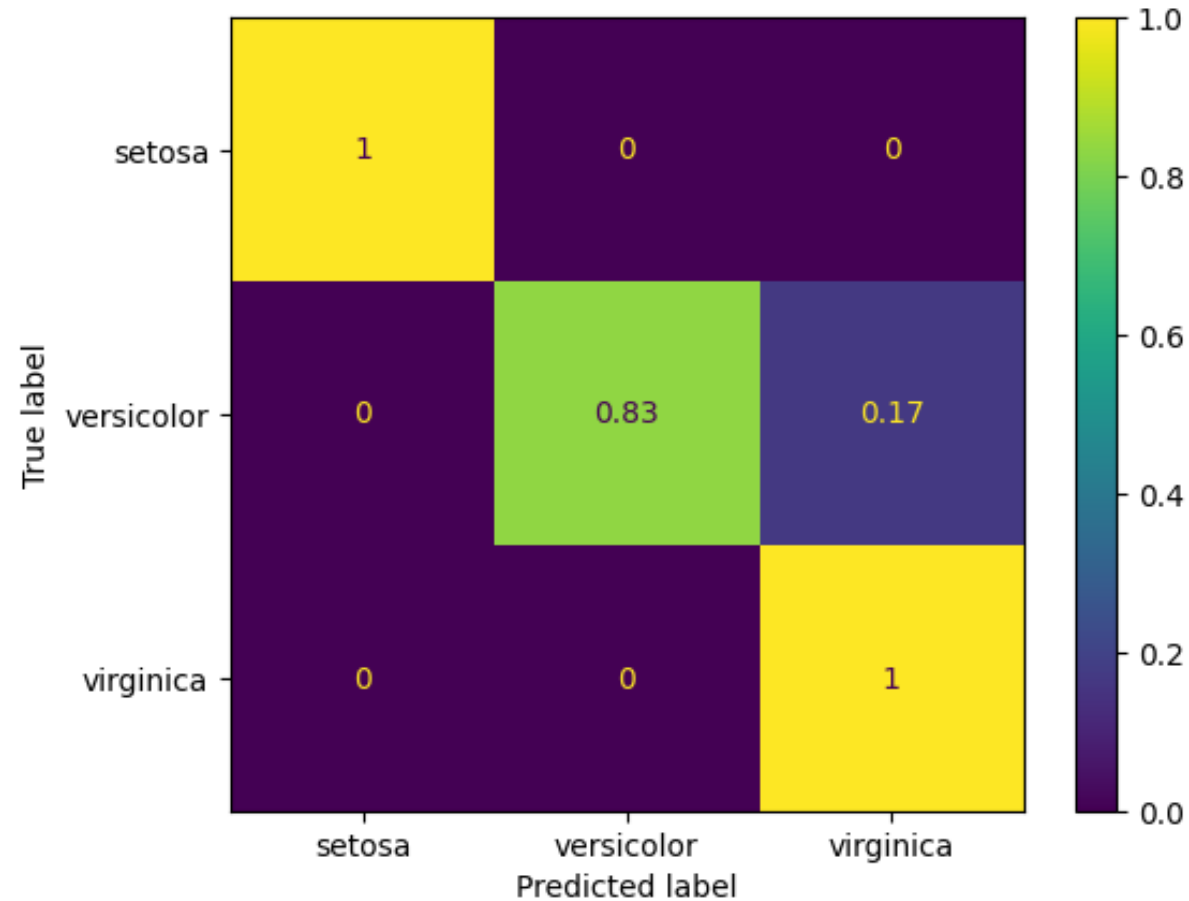
- Create a StandardScaler instance:
`scaler: StandardScaler = StandardScaler()`
- Compute the mean and standard deviation for each column for the scaling in the next step:
`scaler.fit(training_data)`
- Actual standardization:
`training_data_scaled = scaler.transform(training_data)`

Exercise

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Confusion Matrix

`confusion_matrix(y_true, y_predicted)`



Exercise
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